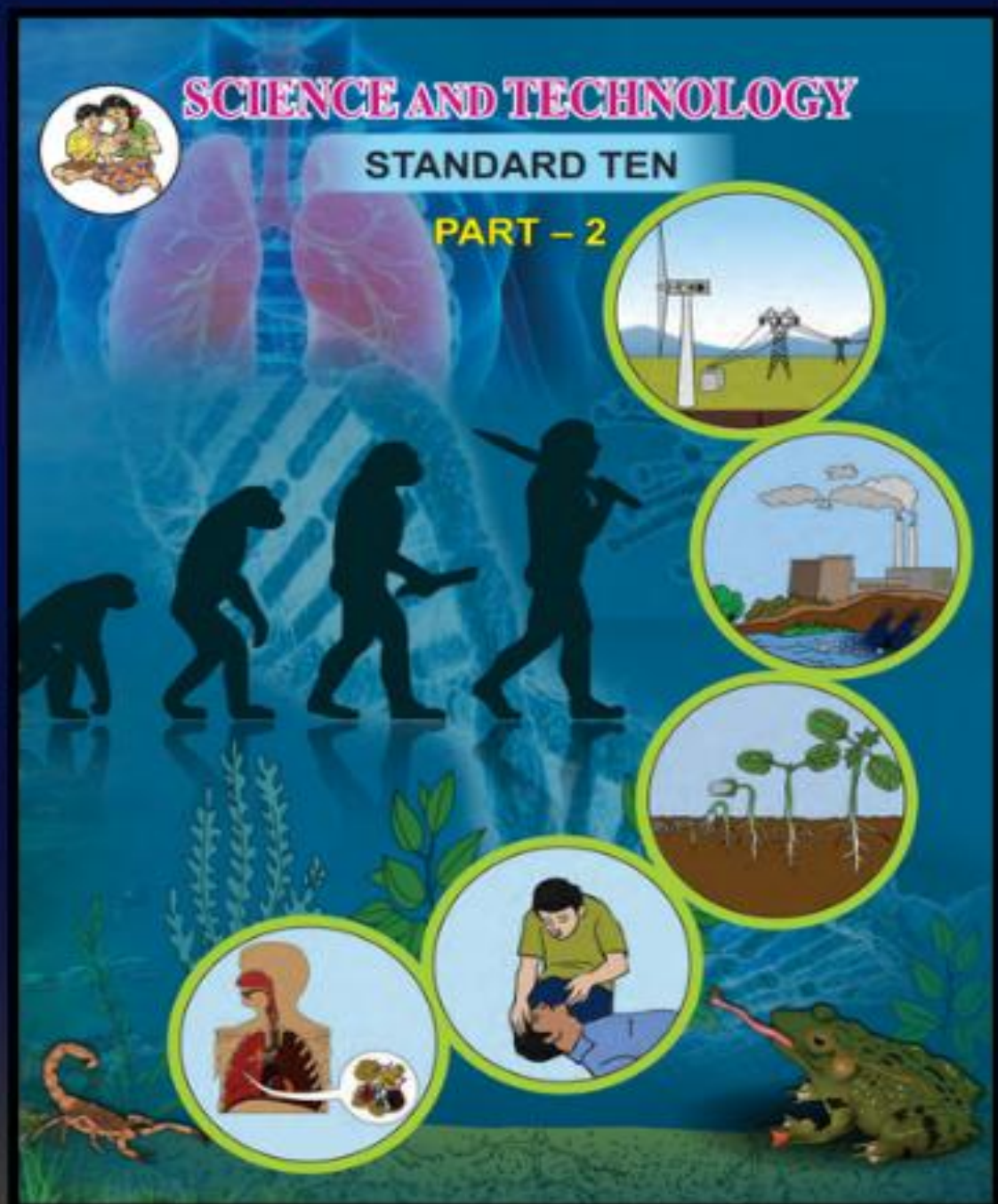


CLASS 10 | MH BOARD

SCIENCE 2 NOTES!



Chapter 1. Heredity and Evolution

Heredity and Hereditary Changes

- **Heredity** is the transfer of biological characters from one generation to another via genes.
- **Gregor Johann Mendel** is considered the pioneer of modern genetics.
- **Hugo de Vries** proposed the theory of mutation in 1901, which explained the reasons for sudden changes.
- In 1902, **Walter Sutton** observed paired chromosomes in grasshopper cells, leading to the discovery that genes are carried on chromosomes.
- In 1944, **Oswald Avery, Mclyn McCarty, and Colin McLeod** proved that, except for viruses, DNA is the genetic material in all living organisms.
- **Francois Jacob and Jack Monod** proposed a model in 1961 for protein synthesis with the help of DNA in bacterial cells.
- The science of heredity is useful for diagnosing and treating hereditary disorders, producing hybrid varieties of plants and animals, and in industrial processes using microbes.

Transcription, Translation and Translocation

- **Central Dogma:** The process by which DNA synthesizes proteins through RNA is called the 'Central Dogma'.
- **Transcription:** The process of RNA synthesis, where mRNA is produced from a DNA strand. The nucleotide sequence in the mRNA is complementary to the DNA strand, but uracil replaces thymine.
- **Triplet Codon:** The message on mRNA contains codes for amino acids. The code for each amino acid consists of three nucleotides.
- **Dr. Har Gobind Khorana**, a scientist of Indian origin, contributed to the discovery of triplet codons for 20 amino acids and was awarded the Nobel Prize in 1968 for this work.
- **Translation:** As per the message on mRNA, tRNA supplies amino acids. The tRNA has an 'anticodon' that is a complementary sequence to the codon on the mRNA. This process is called translation.
- **Translocation:** The ribosome keeps on moving along the mRNA by a distance of one triplet codon at a time. This process is called translocation.

Mutation

- **Mutation** is a sudden change that occurs in genes. It can be caused by a nucleotide of a gene changing its position.
- Mutations can be minor but some may be considerable. For example, a mutation can cause

genetic disorders like **sickle cell anemia**.

Evolution

- **Evolution** is the gradual, slow-going process of change occurring in living organisms over a long duration.
- The formation of new species due to changes in specific characteristics over several generations, in response to natural selection, is called evolution.
- The first living material (protoplasm) is believed to have formed in the ocean, from which unicellular organisms gradually developed into larger, more complex organisms over billions of years.
- **Theory of Evolution:** This theory states that the first living material (protoplasm) was formed in the ocean, leading to the formation of a unicellular organism. Over time, these unicellular organisms evolved into larger and more complex organisms through slow and gradual changes.

Evidences of Evolution

- **Morphological Evidences:** Similarities in the external structures of animals, like the structure of the mouth and eye position, or in plants, like leaf shape and venation, suggest a common origin and ancestors.
- **Anatomical Evidences:** Despite a lack of superficial similarity between structures like a human hand, an ox's foreleg, a whale's flipper, and a bat's patagium, the similarity in the structure of their bones and joints suggests they may have a common ancestor.
- **Vestigial Organs:** These are degenerated or underdeveloped, useless organs in organisms. They are non-functional in some organisms but may be functional in others. Examples in humans include the appendix, ear pinna muscles, wisdom teeth, and the tail-bone (coccyx).
- **Paleontological Evidences:** The study of fossils, which are the preserved remnants or impressions of organisms buried underground, provides important evidence for evolution.
 - **Carbon Dating:** This method, developed by **Willard Libby**, is used to determine the age of fossils by measuring the radioactivity of C-14 and the ratio of C-14 to C-12.
- **Connecting Links:** Some organisms show morphological characteristics of two different groups, linking them together.
 - **Peripatus** is a connecting link between Annelida and Arthropoda, showing characters of both groups.
 - **Duck-billed platypus** lays eggs like reptiles but has mammary glands and hair like mammals.
 - **Lungfish** performs respiration with lungs, even though it's a fish.
- **Embryological Evidences:** A comparative study of the embryonic development stages of various vertebrates shows extreme similarities during the initial stages, which gradually

decrease. This suggests a common origin for these animals.

Darwin's Theory of Natural Selection

- **Charles Darwin** proposed the theory of natural selection, which preaches the **survival of the fittest**.
- He published a book titled '**Origin of Species**' to explain his concept.
- The theory states that organisms reproduce prolifically and compete in a life-threatening manner. Only those with modifications essential for winning the competition survive and reproduce, giving rise to new species.
- **Objections to the theory** include: natural selection isn't the only factor for evolution, it doesn't explain useful vs. useless modifications, and it lacks an explanation for slow vs. abrupt changes.

Lamarckism

- **Jean-Baptiste Lamarck** proposed that morphological changes in organisms are responsible for evolution. The reason for these changes is the **activity or laziness of the organism**.
- He called this the **principle of 'use or disuse of organs'**. For example, the long neck of a giraffe is due to generations of stretching to reach tall plants.
- This theory, also known as the **theory of inheritance of acquired characters**, was disproved because modifications acquired during a lifetime are not transferred to the next generation.

Speciation

- **Speciation** is the formation of new species of plants and animals, which is an effect of evolution.
- A **species** is a group of organisms that can produce fertile individuals through natural reproduction.
- New species can form from earlier ones due to genetic variation, as well as geographical and reproductive changes or isolation.

Human Evolution

- Human evolution began with monkey-like animals that evolved from ancestors similar to modern lemurs about 7 crore years ago.
- The tail of these animals disappeared, and their hands were improved, leading to the evolution of ape-like animals about 4 crore years ago.
- Ape-like animals that remained in Africa evolved into gorillas and chimpanzees about 2.5 crore years ago.
- The first human-like animals with an erect posture evolved about 2 crore years ago.
- The first record of a human-like animal is the **Ramapithecus** ape from North India and East Africa.
- Skilled human beings began to appear about 20 lakh years ago.
- The **Neanderthal man** is considered the first example of a wise-man (*Homo sapiens*), with a brain that had evolved sufficiently about 50,000 years ago.
- The **Cro-Magnon man** evolved about 50,000 years ago, and human evolution accelerated after that.
- Agriculture began about 10,000 years ago, and the art of writing was invented about 5,000 years ago, marking the start of history.

Chapter 2. Life Processes in Living Organisms Part -1

Living Organisms and Life Processes

- For an organism to stay alive, its body's various organ systems (digestive, respiratory, circulatory, excretory, and control systems) must function continuously in a coordinated manner.
- This requires a continuous source of energy.
- The main energy sources are carbohydrates, fats, and lipids. Energy is harvested by the mitochondria in each cell.
- Oxygen is also necessary for energy production, and both foodstuffs and oxygen are transported to cells via the circulatory system.
- Carbohydrates are the primary source of energy, providing 4 Kcal per gram.

Living Organisms and Energy Production

- **Cellular respiration:** The step-by-step oxidation of glucose (a carbohydrate) in cells to obtain energy in the form of ATP.
- There are two main methods of cellular respiration: **aerobic respiration** (involves oxygen) and **anaerobic respiration** (does not involve oxygen).
- Aerobic respiration occurs in three steps:
 1. **Glycolysis:** Occurs in the cytoplasm. A glucose molecule is oxidized to form two molecules each of pyruvic acid, ATP, NADH₂, and water. Pyruvic acid is then converted to Acetyl-Coenzyme-A, releasing NADH₂ and CO₂.
 2. **Tricarboxylic acid cycle (Krebs cycle):** Both Acetyl-CoA molecules enter the mitochondria and are completely oxidized in a cyclic chain of reactions. This process derives molecules of CO₂, H₂O, NADH₂, and FADH₂.
 3. **Electron transfer chain reaction:** NADH₂ and FADH₂ molecules from the previous steps participate in this reaction, which occurs in the mitochondria. This process produces 3 ATP molecules from each NADH₂ and 2 ATP molecules from each FADH₂, along with water molecules.
- **Anaerobic Respiration:** Performed by organisms that cannot live in the presence of oxygen (e.g., many bacteria). It is also performed by some higher plants, animals, and aerobic microorganisms if there is a depletion of oxygen in the surrounding.
 - It consists of **glycolysis and fermentation**.
 - Glucose is incompletely oxidized, producing less energy.

- Pyruvic acid is converted into other organic acids (e.g., lactic acid) or alcohol with the help of some enzymes.
- In our muscle cells, anaerobic respiration during exercise leads to the accumulation of lactic acid, causing us to feel tired.
- **ATP (Adenosine triphosphate):** An energy-rich molecule with energy stored in the bonds by which phosphate groups are attached to each other. It serves as the 'energy currency' of the cell, as energy is derived by breaking its phosphate bond.

Energy from different food components

- **Carbohydrates** are the main source of energy. Excess carbohydrates are stored as glycogen in the liver and muscles.
- **Proteins** are macromolecules made of many amino acids. They provide 4 Kcal of energy per gram. After digestion, amino acids are used by organs and cells to produce various proteins for the body. Excess amino acids are not stored but broken down, and the ammonia is eliminated from the body.
- **Lipids** are substances formed by a chemical bond between fatty acids and alcohol. They provide 9 KCal of energy per gram. Digested lipids are converted into fatty acids and alcohol, which are used to produce various substances like phospholipids for plasma membranes and hormones. Excess lipids are stored in adipose connective tissue.
- **Vitamins** are a group of heterogeneous compounds essential for various body processes. They are fat-soluble (A, D, E, K) or water-soluble (B, C).
- **Water** makes up about 65-70% of the body and is an essential nutrient for the proper functioning of cells and the whole body.
- **Fibers** are not digested by humans but are important for the digestion of other substances and the egestion of undigested materials.

Cell Division: An Essential Life Process

- Cell division is crucial for the formation of a new organism, the growth of a multicellular organism, and the healing of wounds.
- Before any cell division, the cell doubles its chromosome number.
- There are two types of cell division: **mitosis** and **meiosis**.

Mitosis

- Occurs in somatic cells and stem cells.

- It has two main steps: **karyokinesis** (nuclear division) and **cytokinesis** (cytoplasmic division).
- **Karyokinesis** is completed in four stages:
 1. **Prophase:** Chromosomes condense, becoming short and thick. Centrioles duplicate and move to opposite poles. Nuclear membrane and nucleolus start to disappear.
 2. **Metaphase:** The nuclear membrane disappears completely. All chromosomes align parallel to the equatorial plane of the cell. Spindle fibers are formed between the centromeres of each chromosome and both centrioles.
 3. **Anaphase:** Centromeres split, and sister chromatids separate and are pulled to opposite poles by spindle fibers.
 4. **Telophase:** Chromosomes decondense and become thread-like and invisible. A nuclear membrane forms around each set of chromosomes, creating two daughter nuclei.
- **Cytokinesis:** The cytoplasm divides, forming two new daughter cells. In animal cells, a notch forms and deepens, while in plant cells, a cell plate forms along the midline.
- **Importance of Mitosis:** Essential for body growth, wound healing, and the formation of blood cells.

Meiosis

- Occurs in germ cells.
- It is a two-stage process: **Meiosis-I and Meiosis-II.**
- **Meiosis-I:** In this stage, recombination or crossing over occurs between homologous chromosomes, which are then divided into two haploid cells.
- **Meiosis-II:** The two haploid cells from Meiosis-I undergo division by separating the recombined sister chromatids, forming four haploid daughter cells.
- **Importance of Meiosis:** It is the process for gamete production and spore formation. The crossing over of homologous chromosomes results in genetic recombination, making the four daughter cells genetically different from the parent cell and each other.

Chapter 3. Life Processes in Living Organisms Part - 2

Reproduction

- **Reproduction** is the process by which an existing organism produces a new organism of the same species.
- It is an essential process for maintaining the continuity of a species.
- There are two main methods of reproduction: asexual and sexual reproduction.

Asexual Reproduction

- Asexual reproduction is the formation of a new organism from a single parent without the involvement of gametes.
- The new organism is genetically identical to the parent.
- This is a uniparental process and occurs via mitotic cell division.
- **Advantage:** It is a fast process.
- **Disadvantage:** It lacks genetic recombination.

Asexual reproduction in unicellular organisms:

- **Binary Fission:** A parent cell divides to form two similar daughter cells. It occurs in prokaryotes (bacteria) and protists like Amoeba, Paramecium, and Euglena. The axis of division differs among species: Amoeba (simple), Paramecium (transverse), and Euglena (longitudinal).
- **Multiple Fission:** Asexual reproduction in Amoeba and other protists during adverse conditions. The parent cell forms a protective covering called a **cyst**. The nucleus divides repeatedly, followed by cytoplasmic division, to form many amoebulae. The cyst breaks open when conditions are favorable, releasing the new organisms.
- **Budding:** Occurs in unicellular fungi like yeast. A small bulge called a bud forms on the parent cell, one of the daughter nuclei enters the bud, and the bud eventually separates to live independently as a new cell.

Asexual reproduction in multicellular organisms:

- **Fragmentation:** The parent organism's body breaks into many fragments, and each fragment develops into a new, independent organism. This happens in algae like *Spirogyra* and sponges like *Sycon*.
- **Regeneration:** The ability of an organism to regenerate a broken body part. In animals like *Planaria*, if the body breaks into two parts, each part can regenerate the missing part to form two new organisms.
- **Budding:** An outgrowth (bud) forms on the parent organism's body (e.g., *Hydra*) due to repeated cell division. This bud grows into a small, nascent organism that eventually separates from the parent and lives independently.
- **Vegetative Propagation:** Reproduction in plants using vegetative parts like roots, stems, leaves, and buds. Examples include potatoes (using 'eyes' on tubers) and *Bryophyllum* (buds on leaf margins).
- **Spore Formation:** Fungi like *Mucor* have filamentous bodies with sporangia that release spores. When spores land in a moist, warm place, they germinate to form a new fungal colony.

Sexual Reproduction

- Sexual reproduction involves the fusion of two gametes: a haploid male gamete and a haploid female gamete.
- **Gamete formation:** Gametes are formed by meiosis, which reduces the chromosome number by half.
- **Fertilization:** A diploid zygote is formed by the union of the male and female gametes. The zygote then divides by mitosis to form an embryo, which develops into a new individual.
- **Genetic Variation:** The new individual inherits recombined genes from both parents, showing similarities to both but also having unique characters. This genetic diversity helps organisms adapt to changing environments and prevents extinction.

Sexual reproduction in plants:

- The **flower** is the structural unit of sexual reproduction in plants.
- It consists of four whorls:
 - **Essential whorls:** **Androecium** (male whorl, made of stamens) and **Gynoecium** (female whorl, made of carpels).
 - **Accessory whorls:** **Calyx** (sepals) and **Corolla** (petals), which protect the inner whorls.
- **Pollination:** The transfer of pollen grains from the anther to the stigma.

- **Double Fertilization:** A process in plants where one male gamete unites with the egg cell to form a zygote, and a second male gamete unites with two polar nuclei to form the endosperm.
- After fertilization, the ovule develops into a seed, and the ovary develops into a fruit. The zygote uses the stored food in the endosperm to grow into a new plantlet, a process called **seed germination**.

Sexual Reproduction in Human Beings

- Human reproductive systems are controlled by hormones and sex chromosomes (XX for females, XY for males).

Human Male Reproductive System:

- Consists of testes, various ducts (rete testis, vas eferens, epididymis, vas deferens, ejaculatory duct, urinogenital duct), and glands (seminal vesicles, prostate gland, Cowper's glands).
- Sperms are produced in the testes and become mature as they travel through the ducts.
- **Semen** is formed by sperms and the secretions of the various glands.

Human Female Reproductive System:

- Consists of a pair of ovaries, a pair of oviducts, a single uterus, and a vagina.
- An ovum is released from an ovary alternately each month, from puberty until menopause (around age 45-50).
- **Fertilization** is internal in humans, where one sperm fertilizes the only ovum present in the oviduct to form a zygote.
- **Menstrual Cycle:** A natural, repetitive process in the female reproductive system that occurs every 28-30 days. It is controlled by four hormones: follicle stimulating hormone (FSH), luteinizing hormone (LH), estrogen, and progesterone. If the released oocyte is not fertilized, the uterine lining degenerates and is discarded through the vagina, causing bleeding (menstruation) for approximately five days.

Reproductive and Modern Technology

- **In Vitro Fertilization (IVF):** A technique where fertilization is done in a test tube, and

the resulting embryo is implanted in the woman's uterus at an appropriate time.

- **Surrogacy:** Used when a woman cannot implant an embryo in her uterus. Her oocyte is fertilized with her husband's sperm, and the embryo is implanted in the uterus of another woman, a 'surrogate mother'.
- **Sperm Bank/Semen Bank:** A facility where semen from desired men is collected and stored after a thorough medical check-up. It is used in IVF to fertilize the oocyte of a woman whose partner has sperm-related problems.
- **Twins:** Two embryos develop simultaneously, and two offspring are delivered at the same time.
 - **Monozygotic twins:** Formed from a single embryo that splits into two groups of cells early in development. They are genetically identical and have the same gender.
 - **Dizygotic twins:** Formed when two separate oocytes are fertilized by two separate sperms, creating two zygotes. They are genetically different and may be of the same or different genders.

Reproductive Health

- **Health** is a person's state of being physically, mentally, and socially strong.
- There is a lack of awareness regarding reproductive health due to social customs, illiteracy, and shyness.
- **Sexual Diseases:** Diseases like **syphilis** (symptoms include chancre, rash, fever, and inflammation of joints) and **gonorrhoea** (symptoms include painful urination, pus, and inflammation of the urinary tract) can occur on a large scale.

Population Explosion

- **Population explosion** is the excessive growth of a population in a short time.
- It leads to problems like unemployment, decreasing per capita income, and stress on natural resources.
- **Family planning** is the solution to control population growth.

Chapter 4. Environmental Management

Ecosystem - A review

- An **ecosystem** is formed by biotic and abiotic factors and their interactions.
- Producers like plants, herbivores like deer, and predators like lions all play important roles in the ecosystem.
- Organisms present in filthy places, termites, and insects in dung are also important as they are responsible for cleaning the environment.
- The overall balance of the ecosystem is maintained by the continuous operation of natural cycles like the water, carbon, gaseous, and oxygen cycles, as well as various food chains.

Relationship between Environment and Ecosystem

- **Environment** is a broad concept that includes all physical, chemical, and biological factors affecting living organisms.
- It consists of **natural** (air, atmosphere, water, land, living organisms) and **artificial** factors.
- **Ecology** is the science that deals with the study of the interactions between biotic and abiotic factors of the environment.
- The **ecosystem** is the basic functional unit used to study ecology.

Environmental Conservation

- **Environmental pollution** is an unnecessary and unacceptable change in the surrounding environment due to natural events or human activities.
- Causes of pollution include population explosion, fast industrialization, indiscriminate use of natural resources, deforestation, and unplanned urbanization.
- **Radioactive pollution** can be natural (UV, IR radiations) or artificial (X-rays, radiation from atomic plants). Its effects include cancerous ulceration, destroyed body tissues, genetic changes, and adversely affected vision.
- **Need for Conservation:** There should be large-scale participation of people in environmental conservation to address environmental problems.
- Laws have been enacted for environmental conservation in India, such as the **Forest (Conservation) Act, 1980** and the **Environmental (Protection) Act, 1986**.

Biodiversity and Conservation

- **Biodiversity** is the richness of living organisms in nature due to the variety of organisms, ecosystems, and genetic variations within a species.
- **Three levels of biodiversity:**
 1. **Genetic diversity:** Diversity among organisms of the same species. Each human being is different from others.
 2. **Species diversity:** Innumerable species of organisms in nature. It includes various types of plants, animals, and microbes.
 3. **Ecosystem diversity:** Many ecosystems present in each region, each with its own characteristic organisms and abiotic factors.
- **Sacred Groves:** Forests conserved by society (not the government) in the name of God and considered sacred. They are present not only in the Western Ghats of India but also in the entire country.
- **Conservation Methods:**
 - Protecting rare species of organisms.
 - Establishing national parks and sanctuaries.
 - Declaring some regions as 'bioreserves'.
 - Maintaining a record of traditional knowledge.

Classification of Threatened Species

- **Endangered Species:** The number of these organisms has declined or their habitat has shrunk to such an extent that they may become extinct if conservative measures are not implemented. E.g., Lion-tailed monkey, lesser florican.
- **Rare Species:** The number of these organisms is considerably low. Organisms of these species being endemic may become extinct very fast. E.g., Red panda, Musk deer.
- **Vulnerable Species:** The number of these organisms is extremely less and continues to decline. E.g., Tiger, Lion.
- **Indeterminate Species:** These organisms appear to be endangered but there is no definite and substantial information about them due to their behavioral habits like shyness. E.g., Giant squirrel (Shekru).

Chapter 5. Towards Green Energy

Energy and Use of Energy

- Energy is a primary need in modern civilization, along with food, cloth, and shelter.
- We need energy in different forms for diverse types of work, such as mechanical, chemical, sound, light, or heat energy.
- Energy can be converted from one form to another.
- Different energy sources are used to the different forms of energy necessary for us.

Generation of Electrical Energy

- Most electric power plants are based on **Michael Faraday's principle of electromagnetic induction**.
- **Electromagnetic Induction:** When the magnetic field around a conductor changes, a potential difference is generated across the conductor.
- An **electric generator** is a machine based on this principle that generates electrical energy.
- A **turbine** with blades is used to rotate the magnet in the generator. An energy source is needed to run the turbine.

Types of Power Generation Plants

1. **Thermal Energy based Electric Power Plant:**
 - Uses **coal** as fuel.
 - Water is heated in a boiler by burning coal to produce high-temperature, high-pressure steam.
 - This steam rotates a turbine, which drives a generator to produce electricity.
 - **Energy transformation:** Chemical energy in coal → Thermal energy → Kinetic energy in steam → Kinetic energy in turbine → Electrical energy.
 - **Problems:** Air pollution from gases like carbon dioxide, sulphur oxide, and nitrogen oxide. Soot particles are also released, which can cause respiratory problems. Reserves of coal are limited and will deplete in the future.
2. **Nuclear Energy based Power Plant:**
 - Uses the energy from the fission of nuclei of atoms like **Uranium or Plutonium** to generate high-temperature, high-pressure steam.

- The steam rotates a turbine, which drives a generator to produce electricity.
 - **Energy transformation:** Nuclear energy → Thermal energy → Kinetic energy in steam → Kinetic energy in turbine → Electrical energy.
 - **Nuclear fission (chain reaction):** A neutron is bombarded on a Uranium-235 atom, causing it to fission into Barium and Krypton, releasing three neutrons and 200 MeV energy. These three neutrons cause the fission of three other Uranium-235 atoms, and this process continues, which is called a chain reaction.
 - **Problems:** The products after fission are radioactive and emit harmful radiations, making nuclear waste disposal a big challenge. An accident can be very fatal due to the release of harmful radiations.
3. **Power Generation Plant based on Natural Gas:**
- Uses a gas at very high temperature and pressure, generated by the combustion of **natural gas**, to run a gas turbine.
 - **Energy transformation:** Chemical energy in natural gas → Kinetic energy from combustion → Kinetic energy in turbine → Electrical energy.
 - **Advantages:** The efficiency of this type of power plant is higher than that of coal-based plants. Burning natural gas causes less pollution as it does not contain sulphur.

Towards Green Energy

- **Green energy** is electricity generated from sources that are environment-friendly and perpetual.
 - Examples include water reservoirs, wind, sunlight, and biofuels.
1. **Hydroelectric Energy:**
- Uses the potential energy of water stored in a dam.
 - Water flows from the dam to a turbine, and its kinetic energy drives the turbine, which powers a generator.
 - **Energy transformation:** Potential energy in water → Kinetic energy in flowing water → Kinetic energy in turbine → Electrical energy.
 - **Advantages:** No fuel combustion means no air pollution. If there is sufficient water storage, it is possible to generate electricity whenever necessary.
 - **Problems:** The back-water from the dam can submerge villages, towns, forests, and fertile land, leading to rehabilitation problems. The obstruction of the river flow may have an adverse effect on living creatures in the river.
2. **Electric Energy Generation using Wind Energy:**
- Uses the kinetic energy of wind to rotate the blades of a wind turbine.
 - The turbine's axle is connected to an electric generator, producing electricity.
 - **Energy transformation:** Kinetic energy in wind → Kinetic energy in turbine → Electrical energy.
 - **Advantages:** It's a clean energy source.
 - **Limitations:** Requires a high wind velocity, which is not available everywhere, so its

use is limited.

3. Electric Energy Generation using Solar Energy:

- **Solar photovoltaic cell:** Converts solar radiation directly into electrical energy (DC power) through the **photovoltaic effect**. Solar cells are made of a semiconductor like silicon.
- **Series connection:** Increases the total potential difference while the current generated remains the same.
- **Parallel connection:** Increases the total current while the potential difference remains the same.
- **Solar panels and arrays:** Many solar cells are connected in series and parallel to make solar panels that generate the required current and potential difference. Many such panels are connected together to generate electricity on a larger scale.
- **Advantages:** Electricity is generated without any fuel combustion and thus without air pollution.
- **Limitations:** Can only generate electricity during the day-time.
- **Solar Thermal Power Plant:** Converts solar energy into heat energy using reflectors and absorbers. This heat energy is then used to generate steam to drive a turbine and generator.

Chapter 6. Animal Classification

Animal Classification

- Animal classification is the process of forming groups and sub-groups of animals depending upon similarities and differences among them.
- It helps in the convenient study of a vast variety of animals, provides an idea about animal evolution, and helps to understand the relationship and role of animals in nature.

History of Animal Classification

- **Aristotle** was the first to classify animals, using criteria like body size, habits, and habitats. This is known as the **Artificial method**.
- Later, the **Natural system of classification** was developed, based on criteria like body organization, types of cells, and biochemical properties.
- Currently, the **five-kingdom classification system** by **Robert Whittaker** is followed, with all multicellular animals included in Kingdom: Animalia.

Traditional Method of Animal Classification

- Animal Kingdom is divided into two main groups based on the presence or absence of a **notochord**.
- **Non-Chordates**: Animals that do not have a notochord. Their body is not supported by a rod-like notochord. They are classified into ten phyla, including Porifera, Coelenterata, Platyhelminthes, Annelida, Arthropoda, and Mollusca.
- **Chordates**: Animals that have a supporting notochord in their body. They have pharyngeal gill-slits and a dorsal, hollow nerve cord. Their heart is on the ventral side. The phylum Chordata is divided into three subphyla: Urochordata, Cephalochordata, and Vertebrata.

New System of Classification (Based on modern criteria)

- **Criteria for Classification:**
 - **Grades of Organization:**
 - **Protoplasmic grade**: All life functions performed by a single cell (e.g., unicellular organisms).

- **Cellular grade:** Multicellular animals where tissues are not formed (e.g., Phylum Porifera).
- **Cell-tissue grade:** Cells come together to form tissues that perform all the body functions (e.g., Phylum Cnidaria).
- **Tissue-organ grade:** Tissues form organs, but complete organ systems are absent (e.g., flatworms).
- **Organ-system grade:** Different organs join together to form organ systems that perform specific functions (e.g., Crab, Human).
- **Body Symmetry:**
 - **Asymmetrical:** No imaginary axis can divide the body into two equal halves (e.g., some sponges).
 - **Radial symmetry:** An imaginary cut through the central axis from any plane divides the body into two equal halves (e.g., Starfish).
 - **Bilateral symmetry:** Only one imaginary axis can divide the body into two equal halves (e.g., Insects, Humans).
- **Germ Layers:** Layers of cells formed during embryonic development that give rise to different tissues.
 - **Diploblastic:** Animals with two germ layers: endoderm and ectoderm (e.g., Cnidarians).
 - **Triploblastic:** Animals with three germ layers: mesoderm besides endoderm and ectoderm (e.g., most other animals).
- **Body Cavity (Coelom):** The cavity between the body wall and internal organs.
 - **Eucoelomate:** Has a true body cavity formed from the mesoderm (e.g., Annelida and all phyla after it).
 - **Acoelomate:** Lacks a body cavity (e.g., Porifera, Cnidaria, Platyhelminthes).
 - **Pseudocoelomate:** Has a body cavity that is not formed in the same way as a true coelom (e.g., Aschelminthes).
- **Body Segmentation:** The body is divided into small, similar units called segments (e.g., Earthworm from phylum Annelida).

Kingdom Animalia

Phylum Porifera

- These animals are with the simplest body plan and are called 'Sponges'.
- They bear numerous pores called 'Ostia' and 'Oscula' on their body.
- Most of them are marine, and a few are fresh water dwellers. Body is asymmetrical.
- They do not show locomotion and are referred to as sedentary animals.
- Their body is supported by spicules or spongin fibres.
- They reproduce by budding, an asexual method, and/or by a sexual method.
- **Examples:** Sycon, Euspongia (Bath sponge).

Phylum Coelenterata/Cnidaria

- Body of these animals is cylindrical or umbrella-like.
- Most of these animals are marine. Body is radially symmetrical and diploblastic.
- Cnidoblast-bearing tentacles are present around the mouth for capturing prey and for protection.
- **Examples:** Hydra, Adamsia (Sea anemone), Aurelia (Jelly fish), Corals.

Phylum Platyhelminthes

- Body of these animals is slender and flat like a leaf or strip, so they are called 'flatworms'.
- Most of these animals are endoparasites, but a few are free-living and aquatic.
- Body is acoelomate and bilaterally symmetrical.
- They are triploblastic, meaning their body is made up of three germ layers: endoderm, ectoderm, and mesoderm.
- They are hermaphrodite, with male and female reproductive systems present in the same animal body.
- **Examples:** Planaria, Liver fluke, Tapeworm.

Phylum Aschelminthes

- Body is long, thread-like, or cylindrical, so they are called 'round worms'.
- They are either free living or endoparasites.
- Body is triploblastic and pseudocoelomate.
- Body is non-segmented and covered with a tough cuticle.
- They are unisexual.
- **Examples:** Ascaris (Intestinal worm), Filarial worm, Loa loa (Eye worm).

Phylum Annelida

- Body of these animals is long, cylindrical, and metamerically segmented.
- Most are free-living, but a few are ectoparasites.
- These animals are triploblastic, bilaterally symmetrical, and eucoelomate.
- They have setae, parapodia, or suckers for locomotion.

- They are either hermaphrodite or unisexual.
- **Examples:** Earthworm, Leech, Nereis.

Phylum Arthropoda

- The largest phylum, with animals having jointed appendages.
- Found in all types of habitats.
- Body is triploblastic, eucoelomate, bilaterally symmetrical, and segmented.
- Have a chitinous exoskeleton around their body.
- These animals are unisexual.
- **Examples:** Crab, spider, scorpion, millipede, centipede, cockroach, butterfly, honey bee.

Phylum Mollusca

- Second largest phylum in the animal kingdom.
- Body is soft and slimy, hence they are referred to as mollusc.
- Aquatic or terrestrial.
- Body is triploblastic, eucoelomate, non-segmented and soft. Except for animals like snails, their body shows bilateral symmetry.
- Their body is divided into three divisions: head, foot, and visceral mass.
- These animals are unisexual.
- **Examples:** Bivalve, Snail, Octopus.

Phylum Echinodermata

- Calcareous spines are present on the body of these animals, hence they are called as echinoderms.
- Found only in the ocean.
- Their body is triploblastic and eucoelomate. Radially symmetrical in adults but bilaterally symmetrical in larvae.
- They perform locomotion with the help of tube-feet, which are also useful for capturing prey.
- These animals have a good ability for regeneration.
- They are mostly unisexual.
- **Examples:** Star fish, sea-urchin, brittle star, sea-cucumber.

Phylum Hemichordata

- Body of these animals is divided into three parts: proboscis, collar, and trunk.
- Notochord is present only in the proboscis region, hence they are called as hemichordates.
- These animals are also called as 'acorn worms'. They are marine and live in burrows in sand.
- They have one to many pharyngeal gill slits.
- They are unisexual or some may be hermaphrodite.
- Considered a **connecting link** between non-chordates and chordates, as this animal shows the characters of both groups.
- **Examples:** Balanoglossus, Saccoglossus.

Phylum Chordata

- These animals have a supporting notochord in their body.
- The phylum Chordata is classified into three subphyla. Important characters of phylum Chordata are: Notochord is present in the body during at least any developmental stage, Pharyngeal gill slits are present in the body during at least any developmental stage, Single, tubular spinal cord is present on dorsal side of body, Heart is present on ventral side of body.
- Subphylum Vertebrata is divided into six classes:
 - **Class Cyclostomata:** Jaw-less mouth provided with a sucker. Skin is soft and without any scale. Paired appendages are absent. Endoskeleton is cartilaginous. **Example:** Petromyzon, Myxine.
 - **Class Pisces:** Cold-blooded aquatic animals living in marine and fresh waters. Body is spindle shaped to minimize water-resistance. Have paired and unpaired fins for swimming. Exoskeleton is in the form of scales, and endoskeleton is either cartilaginous or bony. Respiration occurs with gills. **Examples:** Rohu, Pomfret, Shark, Electric ray.
 - **Class Amphibia:** Strictly aquatic during larval life (aquatic respiration) but can live on land as adults (aquatic and aerial respiration). Have two pairs of appendages. Exoskeleton is absent. External ear is absent but tympanum is present. **Examples:** Frog, Toad, Salamander.
 - **Class Reptilia:** First true terrestrial animals with creeping movement. Cold-blooded animals. Their skin is dry and scaly. Digits are provided with claws. **Examples:** Tortoise, Lizard, Snake.
 - **Class Aves:** Completely adapted for aerial life. Warm-blooded (Homeotherms). Body is spindle-shaped to minimize air resistance during flight. Forelimbs are modified into wings. Exoskeleton is present in the form of feathers. **Examples:** Peacock, Parrot, Pigeon, Duck, Penguin.
 - **Class Mammalia:** Presence of mammary glands is a typical character. Warm-

blooded. Body is divided into head, neck, trunk, and tail. Exoskeleton is in the form of hairs or fur. **Examples:** Elephant, Human, Kangaroo, Dolphin, Bat.

SVK STUDY

Chapter 7. Introduction to Microbiology

Applied and Industrial Microbiology

- **Applied Microbiology:** The branch of biology where the study of enzymes, proteins, applied genetics, molecular biology, etc., related to some prokaryotic and eukaryotic microbes is performed. This study is used to produce various products like food and medicines on a large scale for society.
- **Industrial Microbiology:** This science is related to the commercial use of microbes, including various economic, social, and environmental processes and products.
- Main features of industrial microbiology include:
 - Production of various products through fermentation, such as bread, cheese, wine, chemicals, enzymes, and medicines.
 - Use of microbes for garbage management and pollution control.

Products

- **Dairy Products:**
 - Since ancient times, milk is converted into products like cheese, butter, cream, kefir, and yoghurt for preservation.
 - Milk is pasteurized and then fermented with lactobacilli. The bacteria convert lactose into lactic acid, which coagulates milk proteins.
 - **Yoghurt:** For industrial production, milk powder is mixed with milk to maintain protein content. Bacterial strains of *Streptococcus thermophilus* and *Lactobacillus delbrueckii* are added to ferment the milk.
 - **Cheese:** Produced on a large scale from cow milk. Microbes like *Lactobacillus lactis* are mixed with milk to impart sourness. An enzyme (rennet or protease from fungi) is used to give a dense texture.
- **Probiotics:** Milk products that contain active, beneficial bacteria such as *Lactobacillus* and *Bifidobacterium*. They maintain the balance of intestinal microorganisms, improve resistance, and help digestion.
- **Bread:** Different types of bread are produced from cereal flour by mixing it with baker's yeast, *Saccharomyces cerevisiae*. The yeast ferments carbohydrates, producing carbon dioxide (CO₂) and ethanol, which makes the bread spongy. Yeast also contains useful contents like proteins, vitamins, and minerals, making the bread nutritive.
- **Vinegar:** Chemically, vinegar is 4% acetic acid (CH₃COOH). It is produced by first fermenting carbon compounds with yeast to obtain ethanol, and then using bacterial strains like *Acetobacter* to microbially degrade ethanol into acetic acid.
- **Microbial Enzymes:** Used in chemical industries instead of chemical catalysts. They are eco-friendly because they are active at low temperatures and pressures, saving energy and

avoiding unnecessary by-products. Examples include oxidoreductases, transferases, and hydrolases.

- **Fuels:**
 - **Methane gas:** A gaseous fuel obtained by microbial anaerobic decomposition of urban, agricultural, and industrial waste.
 - **Ethanol:** A clean, smokeless fuel obtained during fermentation of molasses by the yeast, *Saccharomyces*.
 - **Hydrogen gas:** Considered the fuel of the future, it is released during the biophotolysis of water, a process performed by bacteria.
- **Xanthan Gum:** This gum is obtained by fermentation of starch and molasses with *Xanthomonas* species. It is used for production of pigments, fertilizers, weedicides, and paper.

Microbial Pollution Control

- **Land-filling sites:** Large pits are dug far from residential areas and lined with plastic sheets to prevent soil pollution. Waste is dumped and covered with soil, saw dust, and leafy waste. Microbes present in the soil decompose the waste, forming compost.
- **Sewage Management:** Microbes are used to decompose carbon compounds and destroy pathogens in sewage. They release methane and CO₂. The water released after microbial treatment is environmentally safe.
- **Clean Technology:**
 - **Bioremediation:** Using microbes to decompose man-made chemicals.
 - *Thiobacilli* and *Sulphobacilli* convert metals from low-quality metalloids into compounds before they can leach into the environment.
 - **Hydrocarbonoclastic bacteria (HCB)** like *Pseudomonas spp.* and *Alcanovorax borkumensis* are used to clear oil spills in oceans.
 - Bacteria like *Vibrio* and *Ideonella sakaiensis* can decompose plastic (PET).
 - *Acidophillum spp.* and *Acidobacillus ferrooxidans* can control soil pollution from acid rain by using sulphuric acid as an energy source.
 - *Geobacter* bacteria convert water-soluble uranium salts into insoluble salts, preventing groundwater contamination.

Microbes and Farming

- **Microbial Inoculants:** Produced by fermentation, these inoculants are sprayed on seeds or released into plants to supply nutrients and improve plant growth.
- **Bioinsecticides:** Bacterial and fungal toxins can be integrated into plants using biotechnology to destroy pests. This is beneficial for organic farming.

SVK STUDY

Chapter 8. Cell Biology and Biotechnology

Cell Biology (Cytology)

- **Cytology** is the study of cell structure, types, organelles, and cell division, among other aspects.
- Revolutionary changes have been brought about in the field of human health due to cell biology.
- Research institutes like the **National Center for Cell Science (Pune)** and '**Instem**' (**Bengaluru**) are dedicated to cell research.

Stem Cells

- **Stem cells** are special types of cells present in multicellular organisms that give rise to all other types of cells in the body. They also play an important role in wound healing.
- The new organism formed from a zygote at the earliest stage of development is a mass of cells, which are called stem cells.
- Stem cells are present in the umbilical cord and in the blastocyst stage of embryonic development. In adults, they are found in red bone marrow and adipose connective tissue.
- **Pleuriipotency**: The property of stem cells to be parent cells of all types of human cells.
- **Stem Cell Preservation**: Stem cell samples are collected from sources like cord blood, red bone marrow, or embryo and are kept in small, sterile vials at a very low temperature (-135°C to -190°C) in liquid nitrogen.

Types of Stem Cells

- **Embryonic stem cells**: These are embryonic cells before differentiation starts from the 14th day of conception. These cells can form any type of cell and are the parent cells of all 220 different types of cells in the human body.
- **Adult stem cells**: Stem cells that can be obtained from the body of an adult person. The main sources are red bone marrow, adipose connective tissue, and blood.

Uses of Stem Cells

1. **Regenerative Therapy:**

- **Cell Therapy:** Stem cells are used to replace dead cells in conditions like diabetes, myocardial infarction, Alzheimer's disease, and Parkinson's disease.
- **Blood cell production:** Used to produce blood cells needed for conditions like anemia, thalassemia, and leukemia.

2. **Organ Transplantation:** In case of failure of organs like the kidney and liver, these organs can be produced with the help of stem cells and transplanted.

[Image of Stem cells and organs](#)

Organ Transplantation

- Organs in the human body can become less efficient or non-functional due to various reasons, such as aging, accidents, infections, or disorders.
- **Organ donation** is an important requirement for organ transplantation, as a person's life can be saved if they receive a necessary organ.
- The **Transplantation of Human Organs Act, 1994**, along with its amendments, controls the process of organ and body donation.

Biotechnology

- **Biotechnology** involves bringing about artificial genetic changes and hybridization in organisms for human welfare.
- It includes various branches of science such as cytology, biochemistry, molecular biology, and genetic engineering.
- **Benefits of Biotechnology:**
 - It has made it possible to increase per-hectare crop yield.
 - Expenses on disease control have minimized due to the development of resistant varieties.
 - Yield per annum has increased due to the development of fast fruit-setting varieties.
 - It is possible to develop stress-resistant varieties that can withstand variable temperatures and water-stress.

Commercial Applications of Biotechnology

1. **Crop Biotechnology:**

- **Genetically Modified Crops (GM crops):** Crops with desired characters are developed by integrating a foreign gene into their genome.

- **Bt Cotton:** A gene from the bacterium *Bacillus thuringiensis* is integrated with the cotton genome to produce a toxin that is fatal for bollworms.
- **Golden Rice:** A gene synthesizing Vitamin A (Beta carotene) has been introduced into this variety of rice.
- 2. **Animal Husbandry:** Artificial insemination and embryo transfer are used to improve the quantity and quality of animal products like milk and meat.
- 3. **Human Health:**
 - **Vaccines:** Safer vaccines are now produced by using biotechnology to produce antigens in a laboratory, rather than using killed or semi-killed pathogens.
 - **Gene Therapy:** Used to treat genetic disorders in somatic cells, like Phenylketonuria (PKU).
 - **Cloning:** Production of a replica of any cell, organ, or entire organism. **Reproductive cloning** is the production of a new organism without sperm, and **therapeutic cloning** is the production of stem cells to treat various diseases.
- 4. **DNA fingerprinting:** A technique used to establish the identity of a person based on their unique DNA sequence. It is useful in forensic sciences to identify criminals or establish paternity.
- 5. **Bioremediation:** The use of plants and microorganisms to absorb or destroy toxic chemicals and harmful pollutants. For example, *Pseudomonas* bacteria are useful for cleaning hydrocarbon and oil pollutants from soil and water.

Agricultural Development

- **Green Revolution:** A movement that used improvised dwarf varieties of wheat and rice, proper use of fertilizers, and water management to increase food grain production and save a large population from hunger. Dr. Norman Borlaug (USA) and Dr. M. S. Swaminathan (India) had valuable contributions.
- **White Revolution:** A movement pioneered by Dr. Verghese Kurien that used cooperative movements and biotechnology to make the dairy business a mainstream industry.
- **Blue Revolution:** The production of various useful aquatic organisms with the help of water. The Government of India's 'Nil-Kranti Mission-2016' is an example of this.
- **Organic Farming:** A method of farming that involves a complete ban on chemical fertilizers and pesticides and uses local, sturdy varieties to maintain natural balance.

Chapter 9. Social Health

Social Health

- **Social health** is the ability of a person to establish relationships with other persons.
- An important characteristic of social health is the ability to change one's own behaviour according to changing social conditions.
- Factors important for good social health include: strong personality, having a large number of friends and relatives, proper use of time, trust in others, and respect and acceptance for others.

Factors Endangering Social Health

1. Mental Stress:

- Increased competition for education, employment, and business due to population increase.
- Problems of loneliness and mental stress due to nuclear families and parents working outside.
- Gender inequality and problems like teasing and molestation in adolescent girls.
- Ever-increasing disorder, crime, and violence in society.

2. Addiction:

- Peer-group pressure can lead adolescents to try tobacco, cigarettes, gutkha, alcoholic drinks, and drugs.
- Addictive substances can permanently damage the human nervous system, muscle system, and heart.
- Alcoholism decreases the efficiency of the nervous system (especially the brain) and liver, and it can hinder brain development in adolescents.

3. Communication Media and Modern Technology:

- Excessive and irrational use of modern technology and media is becoming a sensitive issue for social health.
- Continuous use of cell phones can cause physical problems like tiredness, headache, insomnia, and joint pains.
- People who use computers and the internet excessively can become solitary, self-centered, and less sensitive towards others.
- **Selficide:** A person's addiction to taking selfies makes them unaware of their surroundings and the risks, which can prove fatal.

Cyber Crimes

- Cyber crimes are various illegal activities committed with the help of computers and the internet.
- Examples include:
 - Deceiving consumers by showing superior items on websites but selling inferior ones.
 - Hacking into confidential information from government, institutes, and companies.
 - Opening a fake account on social media and using it to tease or exploit people financially.
 - Piracy, which is the illegal sale or misuse of others' copyrighted literature, software, photos, and music.
 - Sending derogatory messages, vulgar pictures, and inflammatory statements through electronic media.
- The **Cyber crime unit** has been newly launched in the police department to investigate and find cyber criminals.
- The **IT Act-2000**, with amendments, was enacted to curb cybercrimes.

Stress Management

- **Laughter clubs:** People relieve mental stress by laughing loudly together.
- **Communication:** Expressing our feelings with friends, family, and teachers helps to relieve stress.
- **Hobbies:** Fostering hobbies like gardening, reading, and cooking helps in properly utilizing free hours and diverting the mind towards positive thinking.
- **Outdoor games:** Provides physical exercise and improves discipline, interaction, and a sense of unity, thereby driving away loneliness and making a person more social.
- **Yoga and meditation:** These practices help in relieving stress, improving concentration, and developing a positive temperament.
- **Counselling:** Medical advice, counselling, and psychotherapy are available for people who face more serious problems like depression and frustration.

Chapter 10. Disaster Management

Disaster

- A **disaster** is a sudden event that leads to a huge loss of life and property.
- Disasters can be **natural** (e.g., floods, earthquakes, famines) or **man-made** (e.g., wars, bomb blasts, accidents).
- Disasters are unpredictable and cause sudden changes in the environment and huge losses to property. They have long-term effects on the economic, social, cultural, and political aspects of life.

The Nature and Extent of Disaster

- Disasters can be of different types and duration, some are short-term and some are long-term.
- Disasters definitely affect the economy of a nation, for example, the destruction of a port can have long-lasting effects due to reconstruction costs.
- Disasters also affect social leadership and can create administrative problems if local governing bodies are impacted.

Disaster Management

- **Disaster management** is either the prevention of disasters or making arrangements to face them or at least achieve the abilities to face them.
- There is a close relationship between disaster management and public participation, which is highly necessary as citizens are the main sufferers in any disaster.
- **Objectives of Disaster Management:**
 - Saving human life and rescuing people during the calamity.
 - Supplying essential commodities to reduce the effects of the disaster.
 - Restoring human life in the region.
 - Rehabilitating disaster victims.
 - Taking protective measures so that such disasters do not recur in the future and to reduce their intensity.

Phases of Disaster Management

1. **Phase of Emergency:** This phase is crucial for saving maximum lives through quick actions like search and rescue operations, medical assistance, and first aid. The gravity of the disaster can be estimated in this phase.
2. **Transitional Phase:** Rehabilitation work starts after the disaster subsides. It includes clearing debris, restoring water supply and roads, and rehabilitating the victims.
3. **Reconstruction Phase:** This is a highly complicated phase that begins during the transitional phase. People reconstruct their buildings and facilities, but it takes a long time for reconstruction to be completed.

Disaster Management Authority

- The **Disaster Management Act, 2005**, has been passed in our country.
- The **National Disaster Management Authority** is chaired by the Prime Minister.
- The **District Disaster Management Authority** is chaired by the District Collector, who is responsible for disaster management and the implementation of rehabilitation schemes at the district level.
- The **National Disaster Response Force (NDRF)** has been established as per the Disaster Management Act, 2005, with divisions working in the army.

First Aid and Emergency Action

- **First aid** is the primary help offered to people injured in disasters before they receive actual medical treatment.
- **Objectives of First Aid:** To relieve the pains, prevent the condition of victims from deteriorating, and save lives.
- A **First Aid Kit** should contain material necessary for first aid, such as bandage strips, wound gauze, antiseptic, and a thermometer.

Mock Drill

- A **mock drill** is a practice to check the preparedness for facing a disaster as early as possible.
- It involves creating a virtual disaster situation to check the reaction time and to identify possible errors and risks.
- Firefighters often arrange mock drills in schools to demonstrate how to extinguish fires and rescue people.
- **Objectives of a Mock Drill:** Evaluating the response to a disaster and improving the coordination between various departments of disaster control.

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